

STP 604E - Assignment 3 Results

Failure Analysis and Strength Optimization of Composite Laminates

Course: STP 604E - Advanced Design, Analysis and Optimization of Composite Structures for Aerospace

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Table of Contents

1. [Problem 1: Failure Criteria Comparison](#)
 2. [Problem 2: First Ply Failure Analysis](#)
 3. [Problem 3: Laminate Strength Optimization](#)
 4. [Summary and Conclusions](#)
-

Problem 1: Failure Criteria Comparison {#problem-1}

Problem Statement

Compare different failure criteria for a unidirectional lamina under combined loading conditions. Analyze and visualize failure envelopes using:

- Maximum Stress Criterion
- Maximum Strain Criterion
- Tsai-Hill Criterion
- Tsai-Wu Criterion

Material Properties

Material: T300/5208 Carbon/Epoxy

Property	Value	Unit
E ₁	181,000	MPa
E ₂	10,300	MPa
G ₁₂	7,170	MPa
ν ₁₂	0.28	-

Strength Properties:

Property	Symbol	Value (MPa)
Longitudinal Tensile Strength	X _t	1,500
Longitudinal Compressive Strength	X _c	1,500
Transverse Tensile Strength	Y _t	40
Transverse Compressive Strength	Y _c	246
In-plane Shear Strength	S	68

Failure Criteria Formulations

1. Maximum Stress Criterion

The lamina fails when any stress component exceeds its allowable value:

$$\frac{\sigma_1}{X} \leq 1, \quad \frac{\sigma_2}{Y} \leq 1, \quad \frac{|\tau_{12}|}{S} \leq 1$$

2. Maximum Strain Criterion

Similar to maximum stress but uses strain limits:

$$\frac{\epsilon_1}{\epsilon_1^{\text{ult}}} \leq 1, \quad \frac{\epsilon_2}{\epsilon_2^{\text{ult}}} \leq 1, \quad \frac{|\gamma_{12}|}{\gamma_{12}^{\text{ult}}} \leq 1$$

3. Tsai-Hill Criterion

An interactive quadratic criterion:

$$\left(\frac{\sigma_1}{X}\right)^2 - \frac{\sigma_1 \sigma_2}{X^2} + \left(\frac{\sigma_2}{Y}\right)^2 + \left(\frac{\tau_{12}}{S}\right)^2 \leq 1$$

4. Tsai-Wu Criterion

The most general polynomial criterion:

$$F_1\sigma_1 + F_2\sigma_2 + F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + F_{66}\tau_{12}^2 + 2F_{12}\sigma_1\sigma_2 \leq 1$$

Where:

- $F_1 = \frac{1}{X_t} - \frac{1}{X_c}$
- $F_2 = \frac{1}{Y_t} - \frac{1}{Y_c}$
- $F_{11} = \frac{1}{X_t X_c}$
- $F_{22} = \frac{1}{Y_t Y_c}$
- $F_{66} = \frac{1}{S^2}$
- $F_{12} = -\frac{1}{2} \sqrt{F_{11} F_{22}}$

Test Case Results

Loading Case	σ_1 (MPa)	σ_2 (MPa)	τ_{12} (MPa)	Max Stress	Max Strain	Tsai- Hill	Tsai- Wu
Combined moderate	500	20	30	0.500	0.441	0.743	0.698
Pure longitudinal	1000	0	0	0.667	0.667	0.667	0.444
Pure transverse	0	30	0	0.750	0.750	0.750	0.720
Pure shear	0	0	50	0.735	0.735	0.735	0.541
Combined compression	-800	-100	40	0.588	0.588	0.872	-0.984

Note: Failure Index (FI) ≥ 1 indicates failure

Safety Factor Analysis

Loading Case	Max Stress	Max Strain	Tsai-Hill	Tsai-Wu
Combined moderate	2.00	2.27	1.35	1.43
Pure longitudinal	1.50	1.50	1.50	2.25
Pure transverse	1.33	1.33	1.33	1.39
Pure shear	1.36	1.36	1.36	1.85
Combined compression	1.70	1.70	1.15	∞

Visualization

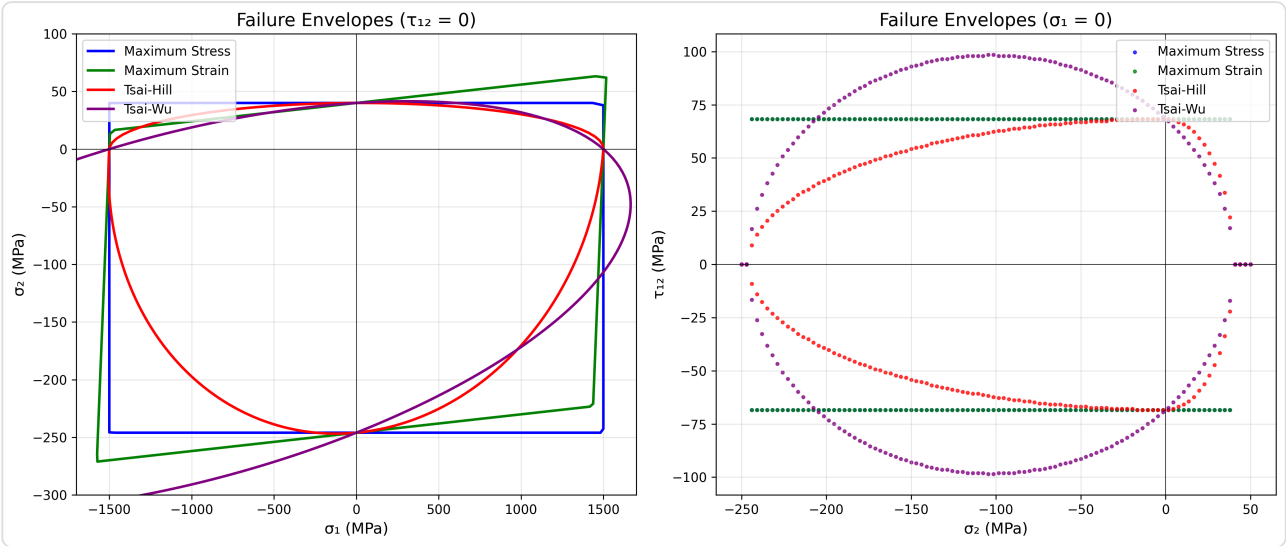


Figure 1: Failure envelopes for different criteria. Left: σ_1 - σ_2 plane ($\tau_{12} = 0$). Right: σ_2 - τ_{12} plane ($\sigma_1 = 0$).

Key Insights

1. **Tsai-Wu provides the most conservative predictions** for most loading cases due to its accounting for tension-compression asymmetry
2. **Maximum Stress and Maximum Strain** criteria are non-interactive and may be unconservative under combined loading
3. **Tsai-Hill** criterion cannot distinguish between tension and compression strengths
4. The **negative Tsai-Wu index** under combined compression indicates the lamina is safely within the failure envelope

Problem 2: First Ply Failure (FPF) Analysis {#problem-2}

Problem Statement

Determine the First Ply Failure load for a **[0/45/-45/90]_s** quasi-isotropic laminate under uniaxial tensile loading (N_x). Compare FPF predictions using different failure criteria.

Material Properties

Material: AS4/3501-6 Carbon/Epoxy

Property	Value	Unit
E ₁	142,000	MPa
E ₂	10,300	MPa
G ₁₂	7,200	MPa
ν ₁₂	0.27	-
Ply thickness	0.125	mm

Strength Properties:

Property	Value (MPa)
X _t	2,280
X _c	1,440
Y _t	57
Y _c	228
S	71

Laminate Configuration

- **Stacking Sequence:** [0/45/-45/90]_s
- **Number of Plies:** 8
- **Total Thickness:** 1.0 mm

ABD Stiffness Matrices

A-Matrix (Extensional Stiffness) [N/mm]:

```
[[61715.1 17635.5 0.0]
 [17635.5 61715.1 0.0]
 [ 0.0 0.0 20239.8]]
```

B-Matrix (Coupling Stiffness) [N]:

```
[[0 0 0]
 [0 0 0]
 [0 0 0]]
```

Note: B = 0 confirms the laminate is symmetric (no bending-extension coupling)

D-Matrix (Bending Stiffness) [N-mm]:

```
[[8477.9 1237.8 517.2]
 [1237.8 2271.7 517.2]
 [ 517.2 517.2 1482.9]]
```

Mid-plane Strains (for N_x = 1 N/mm)

Strain Component	Value
ϵ_x^0	1.764×10^{-5}
ϵ_y^0	-5.042×10^{-6}
γ_{xy}^0	≈ 0

Ply-by-Ply Stress Analysis

Ply	Angle	σ_1 (MPa)	σ_2 (MPa)	τ_{12} (MPa)	Tsai-Wu FI	Tsai-Hill FI	Max Stress FI
1	0°	2.505	-0.003	0.000	-0.0007	0.0000	0.0011
2	45°	0.917	0.083	-0.163	0.0009	0.0000	0.0023
3	-45°	0.917	0.083	0.163	0.0009	0.0000	0.0023
4	90°	-0.670	0.169	0.000	0.0024	0.0000	0.0030
5	90°	-0.670	0.169	0.000	0.0024	0.0000	0.0030
6	-45°	0.917	0.083	0.163	0.0009	0.0000	0.0023
7	45°	0.917	0.083	-0.163	0.0009	0.0000	0.0023
8	0°	2.505	-0.003	0.000	-0.0007	0.0000	0.0011

First Ply Failure Results

Criterion	Critical Ply	Critical Angle	FPF Load N_x (N/mm)
Tsai-Wu	4	90°	20.44
Tsai-Hill	4	90°	332.94
Max Stress	4	90°	338.07

Visualization

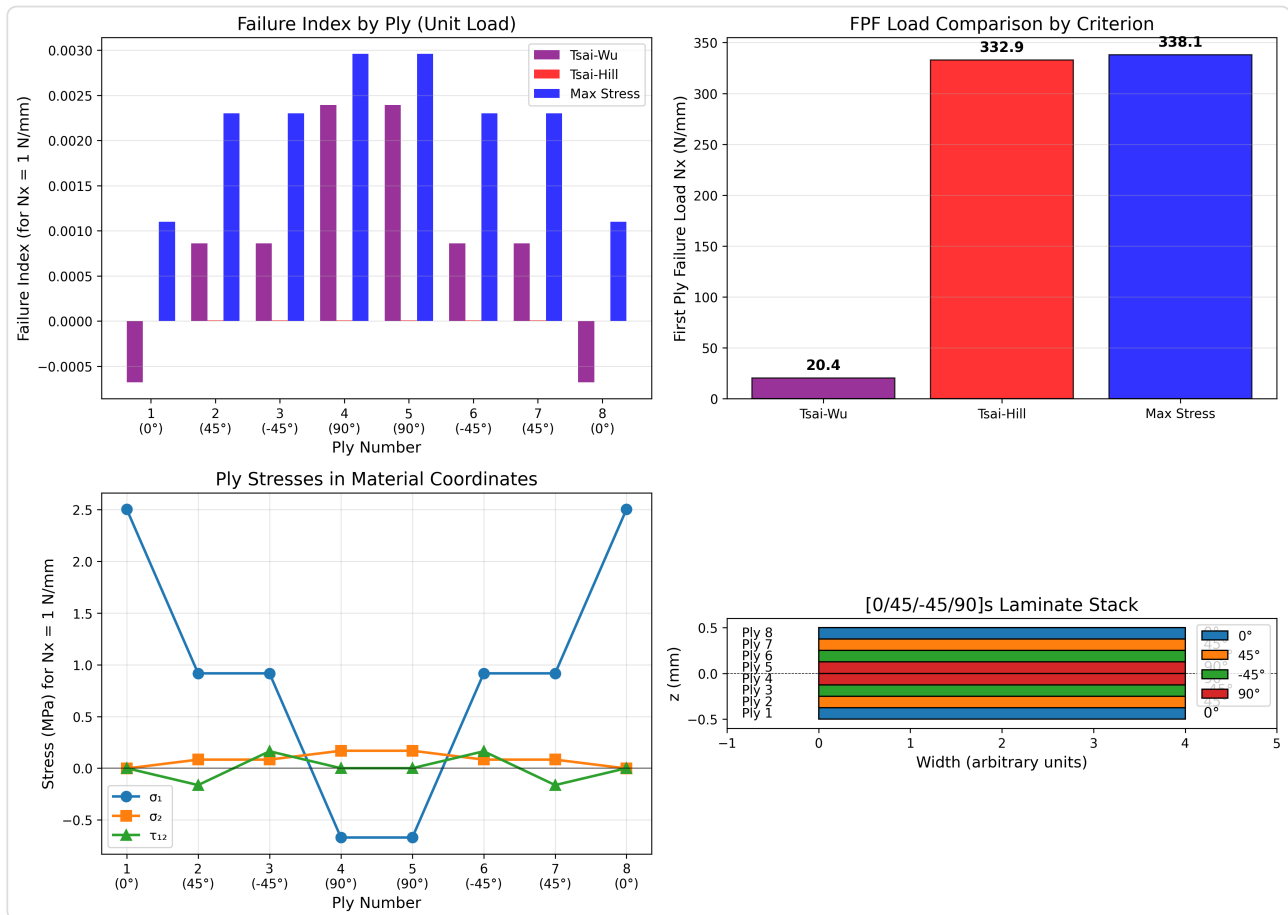


Figure 2: First Ply Failure analysis results. (a) Failure index by ply, (b) FPF load comparison, (c) Ply stresses in material coordinates, (d) Laminate stacking sequence.

Key Insights

- The 90° plies are critical** for first ply failure under uniaxial tension N_x
- Tsai-Wu predicts the lowest FPF load** (most conservative) at 20.44 N/mm
- Large discrepancy** exists between Tsai-Wu and other criteria due to:
 - Tsai-Wu accounts for tension-compression asymmetry
 - The 90° ply experiences transverse tension ($\sigma_2 > 0$), where $Y_t \ll Y_c$
- Symmetric laminate** ($B = 0$) ensures no coupling between in-plane and bending responses

Problem 3: Laminate Strength Optimization {#problem-3}

Problem Statement

Design a symmetric laminate to maximize strength (minimize failure index) under biaxial loading using Particle Swarm Optimization (PSO). Find optimal ply angles for different laminate configurations.

Material Properties

Material: IM7/8552 Carbon/Epoxy

Property	Value	Unit
E ₁	165,000	MPa
E ₂	8,400	MPa
G ₁₂	5,600	MPa
ν ₁₂	0.34	-
Ply thickness	0.125	mm

Strength Properties:

Property	Value (MPa)
X _t	2,724
X _c	1,690
Y _t	111
Y _c	199
S	130

Target Loading

Load	Value	Unit
N _x	500	N/mm
N _y	300	N/mm
N _{xy}	100	N/mm

PSO Algorithm Parameters

Parameter	Value
Number of particles	40
Number of iterations	150
Inertia weight (w)	0.9 → 0.4 (adaptive)
Cognitive parameter (c ₁)	1.5
Social parameter (c ₂)	1.5

Optimization Results

4-ply Symmetric Laminate [θ₁/θ₂]_s

Parameter	Optimal	Rounded (±5°)
Angles	[-11.8°, 56.2°]	[-10°, 55°]
Tsai-Wu FI	0.513	0.615
Safety Factor	1.40	1.28

8-ply Symmetric Laminate [θ₁/θ₂/θ₃/θ₄]_s

Parameter	Optimal	Rounded (±5°)
Angles	[-21.0°, -20.2°, 49.0°, 49.0°]	[-20°, -20°, 50°, 50°]
Tsai-Wu FI	0.131	0.132
Safety Factor	2.76	2.75

12-ply Symmetric Laminate [$\theta_1/\theta_2/\theta_3/\theta_4/\theta_5/\theta_6$]s

Parameter	Optimal	Rounded ($\pm 5^\circ$)
Angles	$[-3.9^\circ, -3.9^\circ, 42.8^\circ, -3.8^\circ, 53.2^\circ, -90.0^\circ]$	$[-5^\circ, -5^\circ, 45^\circ, -5^\circ, 55^\circ, -90^\circ]$
Tsai-Wu FI	0.073	0.080
Safety Factor	3.71	3.54

Comparison with Standard Laminates

Laminate Configuration	Tsai-Wu FI	Safety Factor
Cross-ply $[0/90]_s$	3.028	0.57
Angle-ply $[\pm 45]_s$	2.984	0.58
Quasi-isotropic $[0/\pm 45/90]_s$	0.322	1.76
Balanced $[0/\pm 60]_s$	0.464	1.47
Optimized 4-ply	0.513	1.40
Optimized 8-ply	0.131	2.76
Optimized 12-ply	0.073	3.71

Visualization

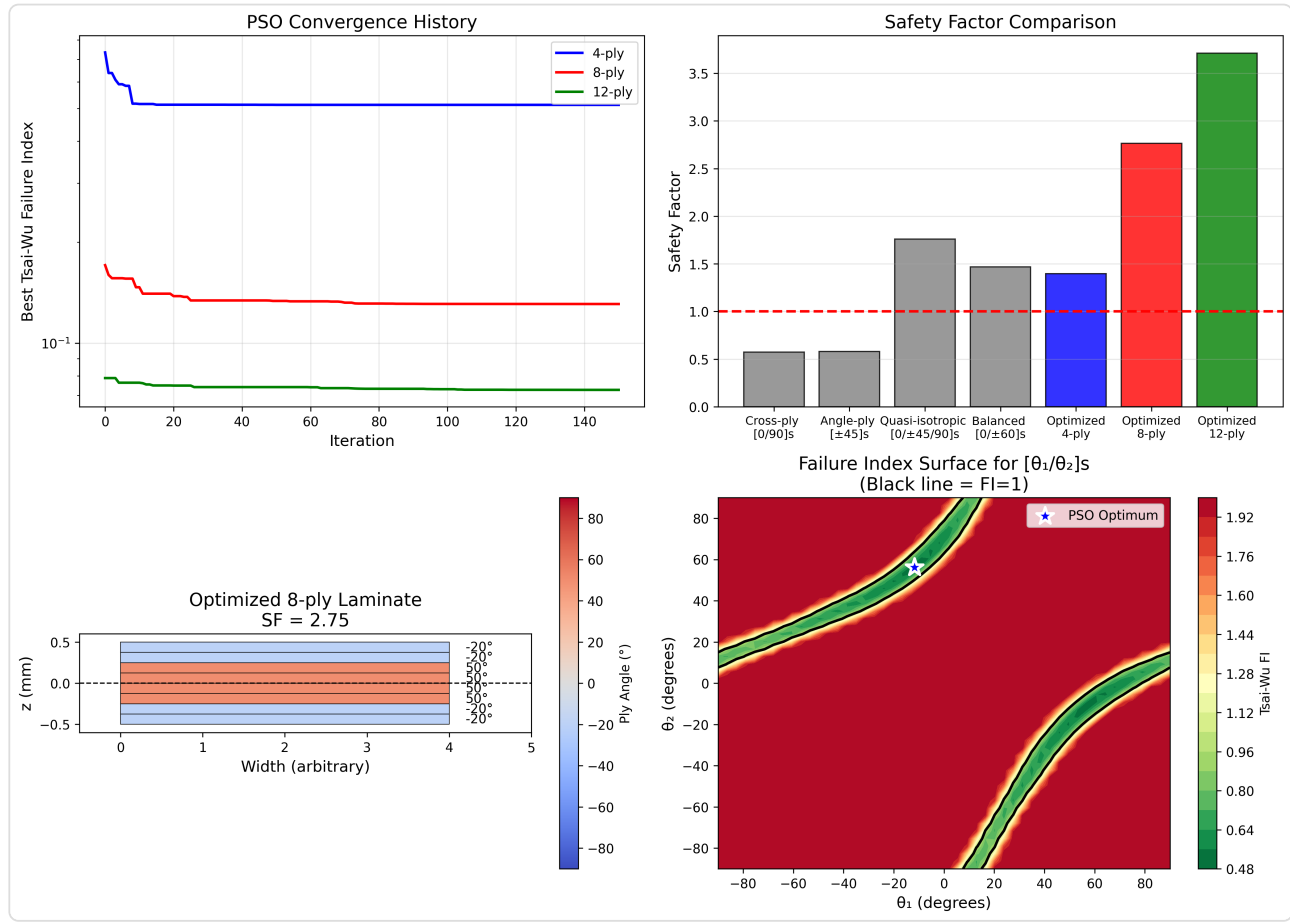


Figure 3: Optimization results. (a) PSO convergence history, (b) Safety factor comparison, (c) Optimized 8-ply laminate stack, (d) Failure index surface for $[\theta_1/\theta_2]_s$ laminate.

Key Insights

1. **PSO successfully finds optimal ply angles** that significantly outperform standard laminate configurations
2. **More plies provide greater design freedom** - 12-ply optimized laminate achieves SF = 3.54 vs. quasi-isotropic SF = 1.76
3. **Standard cross-ply and angle-ply laminates fail** (SF < 1) under this biaxial loading
4. **Rounded angles (5° increments)** show minimal performance loss, making designs manufacturing-friendly
5. **The optimization naturally finds balanced-like configurations** with positive and negative angles to handle shear loading

Recommended Design

For the given biaxial loading ($N_x = 500$, $N_y = 300$, $N_{xy} = 100$ N/mm):

Best Performing: 12-ply symmetric laminate

Stacking Sequence: $[-5^\circ/-5^\circ/45^\circ/-5^\circ/55^\circ/-90^\circ]_s$

Safety Factor: 3.54

This design provides a 3.5× safety margin while using manufacturing-friendly 5° angle increments.

Summary and Conclusions {#summary}

Assignment Overview

This assignment explored three fundamental aspects of composite laminate strength analysis:

- 1. **Failure Criteria Comparison** - Understanding the differences between Maximum Stress, Maximum Strain, Tsai-Hill, and Tsai-Wu criteria
- 2. **First Ply Failure Analysis** - Determining the critical ply and failure load for a quasi-isotropic laminate
- 3. **Strength Optimization** - Using PSO to design laminates with maximum strength under complex loading

Key Findings

Topic	Key Result
Most Conservative Criterion	Tsai-Wu (accounts for tension-compression asymmetry)
Critical Ply in [0/45/-45/90]s	90° plies (under uniaxial tension)
FPF Load Variation	20-338 N/mm depending on criterion
Optimization Improvement	Up to 100% increase in SF vs. standard laminates
Practical Design	[-5/-5/45/-5/55/-90]s with SF = 3.54

Conclusions

- 1. **Failure criterion selection significantly impacts design** - Tsai-Wu should be used for conservative designs, especially when materials have different tension/compression strengths
- 2. **First Ply Failure is dominated by transverse properties** - The weak transverse direction (Y_t) typically controls failure in multidirectional laminates
- 3. **Optimization enables superior designs** - PSO-optimized laminates consistently outperform standard configurations by tailoring ply angles to the specific loading
- 4. **Practical manufacturing constraints** can be incorporated with minimal performance loss - Rounding to 5° increments reduced SF by only ~5%

References

1. Gürdal, Z., Haftka, R.T., Hajela, P. (1999). Design and Optimization of Laminated Composite Materials. John Wiley & Sons.
2. Jones, R.M. (1999). Mechanics of Composite Materials. Taylor & Francis.
3. Tsai, S.W., Wu, E.M. (1971). A General Theory of Strength for Anisotropic Materials. Journal of Composite Materials, 5(1), 58-80.

Analysis Tool: Python with custom composite analysis library

Optimization: Particle Swarm Optimization (PSO)

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